

Topics to discuss

- Taylor series Method
- Numerical Problem
- Homework Problem

YouTube @StartPracticing

Taylor Series Method :

If $y(x)$ is known at a point x , then its value at a nearby point $x+h$ can be obtained using Taylor series expansion

Taylor series Formula :

$$y(x+h) = y(x) + h \frac{dy}{dx} + \frac{h^2}{2!} \frac{d^2y}{dx^2} + \frac{h^3}{3!} \frac{d^3y}{dx^3} + \dots$$

OR

$$= y_0 + h y'_0 + \frac{h^2}{2!} y''_0 + \frac{h^3}{3!} y'''_0 + \dots$$

How to find h :

$$h = x_1 - x_0 \quad ; \quad h = \text{step size}$$

x_0 = initial value

x_1 = next value where solution is required

Case I : When 'h' is given

→ No problem

Case II : When 'h' is NOT given

eg: find $y(0.2)$, given $y(0)=1$

$$\rightarrow h = 0.2 - 0 = 0.2$$

Case III : Multiple steps

eg: find $y(0.3)$, given $y(0)=1$ & $h=0.1$

→ step 1 : $y(0.1)$

→ step 2 : $y(0.2)$

→ step 3 : $y(0.3)$

Numerical Problem :

Q:- Use Taylor method recursively to solve the equation

$$y' = x^2 + y^2, \quad y(0) = 0$$

for the interval $(0, 0.4)$ using two subintervals of 0.2

Solution :

The derivatives of y are given by

$$y' = x^2 + y^2$$

$$y'' = 2x + 2yy'$$

$$y''' = 2 + 2yy'' + 2(y')^2$$

$$\begin{aligned} y^{(4)} &= 2yy''' + 2y'y'' + 4y'y'' \\ &= 6y'y'' + 2yy''' \end{aligned}$$

Iteration 1 :

$$y_1 = y_0 + h y'_0 + \frac{h^2}{2!} y''_0 + \frac{h^3}{3!} y'''_0 + \frac{h^4}{4!} y^{(4)}_0 + \dots$$

we have,

$$h = 0.2, \quad y_0 = 0, \quad x_0 = 0$$

$$\therefore y'_0 = x_0^2 + y_0^2 = 0$$

$$y''_0 = 2x_0 + 2y_0 y'_0 = 0$$

$$y'''_0 = 2 + 2(y'_0)^2 + 2y_0 y''_0 = 2$$

$$y^{(4)}_0 = 6y'_0 y''_0 + 2y_0 y'''_0 = 0$$

Therefore,

$$y_1 = y_0 + h y'_0 + \frac{h^2}{2!} y''_0 + \frac{h^3}{3!} y'''_0 + \frac{h^4}{4!} y^{(4)}_0 + \dots$$
$$= 0 + 0 + 0 + \frac{0.2^3}{3!} \times 2 + 0$$

$$y_1 = 0.002667 \quad \text{at } x = 0.2$$

Iteration 2:

$$x_1 = 0.2, \quad y_1 = 0.002667$$

$$y'_1 = x_1^2 + y_1^2 = 0.2^2 + 0.002667^2 = 0.04$$

$$y''_1 = 2x_1 + 2y_1 y'_1 = 2 \times 0.2 + 2 \times 0.002667 \times 0.04 = 0.400213$$

$$y'''_1 = 2 + 2y_1 y''_1 + 2(y'_1)^2 = 2 + 2 \times 0.002667 \times 0.400213 + 2 \times 0.04^2$$
$$= 2.005335$$

$$y_1^4 = 6y_1' y_1'' + 2y_1 y_1'''$$

$$= 6 \times 0.04 \times 0.400213 + 2 \times 0.002667 \times 2.005335$$

$$= 0.106748$$

$$\begin{aligned} \therefore y_2 &= y_1 + h y_1' + \frac{h^2}{2!} y_1'' + \frac{h^3}{3!} y_1''' + \frac{h^4}{4!} y_1^4 + \dots \\ &= 0.002667 + 0.04 \times 0.2 + \frac{0.400213}{2} \times 0.2^2 \\ &\quad + \frac{2.005335}{6} \times 0.2^3 + \frac{0.2^4}{24} \times 0.106748 \end{aligned}$$

$$= 0.021352$$

$$\therefore y(0.4) = 0.021352$$

Ans

If we use, $h = 0.4$ directly in first iteration
with subdividing, we obtain

$$y(0.4) = 0.021333$$

YouTube @StartPracticing

Homework Problem:

Q: Evaluate $y(4.4)$, given $5xy' = 2 - y^2$, $y(4) = 1$
by Taylor series Method.

Solution:

We have, $5xy' = 2 - y^2$

$$\Rightarrow y' = \frac{2 - y^2}{5x} \quad \text{--- (1)}$$

differentiating both side w.r.t x , we get

$$5xy'' + 5y' = -2yy'$$

$$\therefore y'' = \frac{-2yy' - 5y'}{5x} \quad \text{--- (2)}$$

Again differentiating both side w.r.t x , we get

$$5xy'' + 5y' = -2yy'$$

$$\Rightarrow 5xy''' + 5y'' + 5y'' = -2yy'' - 2y'^2$$

$$\text{or, } y''' = \frac{-2yy'' - 2y'^2 - 10y''}{5x} \quad \text{--- (3)}$$

Also, we have, $y(4) = 1$

$$\text{Let, } x_0 = 4$$

$$y_0 = 1$$

$$\text{and } h = 0.2$$

$$x_1 = x_0 + h$$

$$x_1 = 4.2$$

$$\therefore y(x_1) = y_1$$

According to Taylor series method,

$$y_1 = y_0 + h y'_0 + \frac{h^2}{2!} y''_0 + \frac{h^3}{3!} y'''_0 + \dots$$

Here, to find y_1 , we have to calculate $y'_0, y''_0, y'''_0 \dots$

So,

$$y'_0 = \frac{2 - y_0^2}{5x_0} = \frac{2 - 1}{5 \times 4} = 0.05$$

$$y''_0 = \frac{-2y_0 y'_0 - 5y'_0}{5x_0} = \frac{-2 \times 1 \times 0.05 - 5 \times 0.05}{5 \times 4} = -0.0175$$

$$y'''_0 = \frac{-2y_0 y''_0 - 2y_0'^2 - 10y''_0}{5x_0} = \frac{-2 \times 1 \times (-0.0175) - 2 \times 0.05^2 - 10(-0.0175)}{5 \times 4} \\ = 0.01025$$

Now,

$$\begin{aligned} y_1 &= y_0 + h y'_0 + \frac{h^2}{2!} y''_0 + \frac{h^3}{3!} y'''_0 + \dots \\ &= 1 + 0.2 \times 0.05 + \frac{0.2^2}{2} \times (-0.0175) + \frac{0.2^3}{6} \times 0.01025 \\ &= 1 + 0.01 - 0.00035 + 0.0000013667 \\ &= 1.0096636 \end{aligned}$$

$$\text{So, } y_1 = y(x_1) = y(x_0 + h) = y(4.2)$$

Iteration 2:

$$\text{Now, } x_1 = 4.2$$

$$y_1 = 1.0096636$$

$$h = 0.2$$

$$y_2 = y(x_2)$$

$$\begin{aligned} x_2 &= x_1 + h \\ &= 4.4 \end{aligned}$$

we have to find $y'_1, y''_1, y'''_1 \dots$

$$y_1' = \frac{2 - y_1^2}{5x_1} = \frac{2 - 1.0096636^2}{5 \times 4.2} = 0.04669425$$

$$y_1'' = \frac{-2y_1 y_1' - 5y_1'}{5x_1} = \frac{-2 \times 1.0096636 \times 0.04669425 - 5 \times 0.04669425}{5 \times 4.2}$$

$$= -0.015607725$$

$$y_1''' = \frac{-2y_1 y_1'' - 2y_1'^2 - 10y_1''}{5x_1}$$

$$= \frac{-2 \times 1.0096636 \times (-0.015607725) - 2 \times (0.04669425)^2 - 10 \times (-0.015607)}{5 \times 4.2}$$

$$= 0.484915888$$

$$y_2 = y_1 + h y_1' + \frac{h^2}{2!} y_1'' + \frac{h^3}{3!} y_1''' + \dots$$

$$= 1.0096636 + (0.2 \times 0.04669425) + \frac{0.2^2}{2} \times (-0.015607725) \\ + \frac{0.2^3}{6} \times 0.484915888$$

$$= 1.019002450 - 0.000312155 + 0.000646555$$

$$= 1.019336$$

$$\therefore y_2 = y(x_2) = y(4.4) = 1.019336$$

Ans.

Connect with me



[Start Practicing](#)



[i._am._arfin](#)



[Arfin Parween](#)



[Arfin Parween \(Subscribe here too\)](#)



[Arfin Parween](#)



[Arfin Parween](#)